

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Camera
Manufacturer: 1 Beyond
Firmware Version: N/A
Model(s): PTZ-IP12, PTZ-IP20

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.10.0000-b007	2.6.0

Version History

Module Version	Date	Notes
1_0_0_0	8/26/2020	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`
- DeviceID variable must be set accordingly. Default value is '1'. DeviceID ranges from 1 to '7'.
Example: `InterfaceName.DeviceID = '1'`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='PTZ-IP12')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='PTZ-IP12')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 5500, Model='PTZ-IP12')</code>

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AutoExposure	Value 'Full Auto' 'Iris Priority'	Value 'Manual' 'Bright'	Value 'Shutter Priority'
# AutoExposure example InterfaceName.Set('AutoExposure', 'Full Auto')			
Command AutoFocus	Value 'On'	Value 'Off'	
# AutoFocus example InterfaceName.Set('AutoFocus', 'On')			
Command Backlight	Value 'On'	Value 'Off'	
# Backlight example InterfaceName.Set('Backlight', 'On')			
Command Focus	Value 'Far'	Value 'Near'	Value 'Stop'
Qualifier Key 'Speed'	Qualifier Value 0 to 7 in steps of 1		
# Focus example InterfaceName.Set('Focus', 'Far', {'Speed': 7})			
Command Gain	Value 'Up'	Value 'Down'	Value 'Reset'
# Gain example InterfaceName.Set('Gain', 'Up')			
Command Iris	Value 'Up'	Value 'Down'	Value 'Reset'
# Iris example InterfaceName.Set('Iris', 'Up')			
Command PanTilt	Value 'Up' 'Right' 'Down Left' 'Home'	Value 'Down' 'Up Left' 'Down Right' 'Reset'	Value 'Left' 'Up Right' 'Stop'
Qualifier Key 'Pan Speed'	Qualifier Value 1 to 24 in steps of 1		
Qualifier Key 'Tilt Speed'	Qualifier Value 1 to 20 in steps of 1		
# PanTilt example InterfaceName.Set('PanTilt', 'Up', {'Pan Speed': 24, 'Tilt Speed': 20})			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			
Command Preset	Value 0 – 255		
Qualifier Key	Qualifier Value	Qualifier Value	Qualifier Value

Global Scripter Module Communication Sheet

'Action'	'Reset'	'Save'	'Recall'
# Preset example InterfaceName.Set('Preset', 255, {'Action': 'Reset'})			
Command Shutter	Value 'Up'	Value 'Down'	Value 'Reset'
# Shutter example InterfaceName.Set('Shutter', 'Up')			
Command WhiteBalance	Value 'Auto' 'One Push'	Value 'Indoor' 'Manual'	Value 'Outdoor' 'One Push Trigger'
# WhiteBalance example InterfaceName.Set('WhiteBalance', 'Auto')			
Command Zoom	Value 'Tele'	Value 'Wide'	Value 'Stop'
Qualifier Key 'Speed'	Qualifier Value 0 to 7 in steps of 1		
# Zoom example InterfaceName.Set('Zoom', 'Tele', {'Speed': 7})			

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},
```

FeedbackHandler)

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command AutoExposure	Value 'Full Auto' 'Iris Priority'	Value 'Manual' 'Bright'	Value 'Shutter Priority'
# AutoExposure example InterfaceName.Update('AutoExposure') Value = InterfaceName.ReadStatus('AutoExposure') InterfaceName.SubscribeStatus('AutoExposure', None, FeedbackHandler)			
Command AutoFocus	Value 'On'	Value 'Off'	
# AutoFocus example InterfaceName.Update('AutoFocus') Value = InterfaceName.ReadStatus('AutoFocus') InterfaceName.SubscribeStatus('AutoFocus', None, FeedbackHandler)			
Command Backlight	Value 'On'	Value 'Off'	
# Backlight example InterfaceName.Update('Backlight') Value = InterfaceName.ReadStatus('Backlight') InterfaceName.SubscribeStatus('Backlight', None, FeedbackHandler)			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
# ConnectionStatus example Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command Power	Value 'On'	Value 'Off'	Value 'Internal Power Circuit Error'
# Power example InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)			
Command WhiteBalance	Value 'Auto' 'One Push'	Value 'Indoor' 'Manual'	Value 'Outdoor' 'One Push Trigger'
# WhiteBalance example InterfaceName.Update('WhiteBalance') Value = InterfaceName.ReadStatus('WhiteBalance') InterfaceName.SubscribeStatus('WhiteBalance', None, FeedbackHandler)			

Cable and Adapter Requirements

Captive Screw to 8 pin DIN Serial Cable

Notes for the Device

To ensure the functionality of the driver, please set the following:

- Menu → System → Protocol → VISCA
- On the bottom of the camera, set switch 1 and 2 to the OFF position for the SW2 dip switches

Serial communication

Port Type: RS-232, RS-485

Baud Rate: 9600

Data Bits: 8

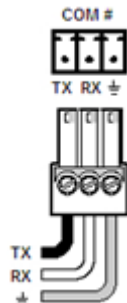
Parity: None

Stop Bits: One

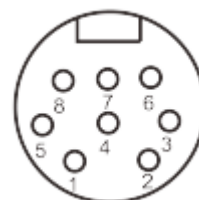
Flow Control: None

Pin Assignments Diagram

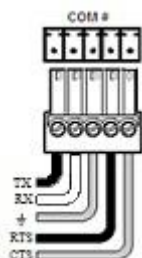
RS-232



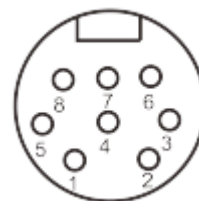
Signal	Main Cable	Pin	Signal
TxD	↗	3	TxD
RxD	↖	5	RxD
GND	—	4/6	GND



RS-485



Please contact the manufacturer for pin diagram.



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scriptor ethernet interface.

Port Type:	Ethernet
Device Listen Port:	5500
Device Reply Port:	5500
Logon Credentials Supported:	No
Multi-Connection Capabilities:	Undetermined
Listen Port Changeability:	No
Reply Port Changeability:	Yes

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Only the **Default Reply Port** is changeable.

To ensure the functionality of the driver, please set the following:

- Menu → System → Protocol → VISCA
- On the bottom of the camera, set switch 1 and 2 to the OFF position for the SW2 dip switches

Appendix A. Set Commands

Auto Exposure Bright	\x81\x01\x049\x0D\xff
Auto Exposure Full Auto	\x81\x01\x049\x00\xff
Auto Exposure Iris Priority	\x81\x01\x049\x0B\xff
Auto Exposure Manual	\x81\x01\x049\x03\xff
Auto Exposure Shutter Priority	\x81\x01\x049\x0A\xff
Auto Focus Off	\x81\x01\x048\x03\xff
Auto Focus On	\x81\x01\x048\x02\xff
Backlight Off	\x81\x01\x043\x03\xff
Backlight On	\x81\x01\x043\x02\xff
Focus Far Speed 0	\x81\x01\x04\x08\xff
Focus Far Speed 7	\x81\x01\x04\x08'\xff
Focus Near Speed 0	\x81\x01\x04\x080\xff
Focus Near Speed 7	\x81\x01\x04\x087\xff
Focus Stop Speed 0	\x81\x01\x04\x08\x00\xff
Focus Stop Speed 7	\x81\x01\x04\x08\x00\xff
Gain Down	\x81\x01\x04\x0C\x03\xff
Gain Reset	\x81\x01\x04\x0C\x00\xff
Gain Up	\x81\x01\x04\x0C\x02\xff
Iris Down	\x81\x01\x04\x0B\x03\xff
Iris Reset	\x81\x01\x04\x0B\x00\xff
Iris Up	\x81\x01\x04\x0B\x02\xff
Pan Tilt Down Left Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x02\xff
Pan Tilt Down Left Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x01\x02\xff
Pan Tilt Down Left Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x01\x02\xff
Pan Tilt Down Left Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x01\x02\xff
Pan Tilt Down Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x03\x02\xff
Pan Tilt Down Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x03\x02\xff
Pan Tilt Down Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x03\x02\xff
Pan Tilt Down Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x03\x02\xff
Pan Tilt Down Right Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x02\x02\xff
Pan Tilt Down Right Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x02\x02\xff
Pan Tilt Down Right Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x02\x02\xff
Pan Tilt Down Right Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x02\x02\xff
Pan Tilt Home Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x04\xff
Pan Tilt Home Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x04\xff
Pan Tilt Home Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x04\xff
Pan Tilt Home Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x04\xff
Pan Tilt Left Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x03\xff
Pan Tilt Left Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x01\x03\xff
Pan Tilt Left Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x01\x03\xff
Pan Tilt Left Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x01\x03\xff
Pan Tilt Reset Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x05\xff

Global Scriptor Module Communication Sheet

Revision: 8/26/2020

Pan Tilt Reset Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x05\xff
Pan Tilt Reset Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x05\xff
Pan Tilt Reset Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x05\xff
Pan Tilt Right Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x02\x03\xff
Pan Tilt Right Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x02\x03\xff
Pan Tilt Right Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x02\x03\xff
Pan Tilt Right Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x02\x03\xff
Pan Tilt Stop Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x03\x03\xff
Pan Tilt Stop Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x03\x03\xff
Pan Tilt Stop Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x03\x03\xff
Pan Tilt Stop Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x03\x03\xff
Pan Tilt Up Left Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x01\xff
Pan Tilt Up Left Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x01\x01\xff
Pan Tilt Up Left Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x01\xff
Pan Tilt Up Left Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x01\x01\xff
Pan Tilt Up Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x03\x01\xff
Pan Tilt Up Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x03\x01\xff
Pan Tilt Up Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x03\x01\xff
Pan Tilt Up Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x03\x01\xff
Pan Tilt Up Right Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x02\x01\xff
Pan Tilt Up Right Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x02\x01\xff
Pan Tilt Up Right Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x02\x01\xff
Pan Tilt Up Right Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x02\x01\xff
Power Off	\x81\x01\x04\x00\x03\xff
Power On	\x81\x01\x04\x00\x02\xff
Preset 0 Action Recall	\x81\x01\x04?\x02\x00\xff
Preset 0 Action Reset	\x81\x01\x04?\x00\x00\xff
Preset 0 Action Save	\x81\x01\x04?\x01\x00\xff
Preset 255 Action Recall	\x81\x01\x04?\x02\xff\xff
Preset 255 Action Reset	\x81\x01\x04?\x00\xff\xff
Preset 255 Action Save	\x81\x01\x04?\x01\xff\xff
Shutter Down	\x81\x01\x04\x0A\x03\xff
Shutter Reset	\x81\x01\x04\x0A\x00\xff
Shutter Up	\x81\x01\x04\x0A\x02\xff
White Balance Auto	\x81\x01\x045\x00\xff
White Balance Indoor	\x81\x01\x045\x01\xff
White Balance Manual	\x81\x01\x045\x05\xff
White Balance One Push	\x81\x01\x045\x03\xff
White Balance One Push Trigger	\x81\x01\x04\x10\x05\xff
White Balance Outdoor	\x81\x01\x045\x02\xff
Zoom Stop Speed 0	\x81\x01\x04\x07\x00\xff
Zoom Stop Speed 7	\x81\x01\x04\x07\x00\xff
Zoom Tele Speed 0	\x81\x01\x04\x07 \xff
Zoom Tele Speed 7	\x81\x01\x04\x07 \xff

Global Scripter Module Communication Sheet

Zoom Wide Speed 0	\x81\x01\x04\x070\xff
Zoom Wide Speed 7	\x81\x01\x04\x070\xff

Appendix B. Update Commands

Auto Exposure	\x81\x09\x049\xff
Auto Focus	\x81\x09\x048\xff
Backlight	\x81\x09\x043\xff
Power	\x81\x09\x04\x00\xff
White Balance	\x81\x09\x045\xff